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New Perspectives on Risk Factors for Exertional Heat Stroke

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Intense exercise in extreme heat can increase the risk of developing exertional heat stroke (EHS). EHS is 100% survivable with appropriate care, and it is imperative that health care professionals recognize predisposing factors that may increase susceptibility to EHS. Understanding risk factors for EHS will enable clinicians to create effective prevention strategies to improve exercise heat tolerance and mitigate EHS risk. This review addresses new perspectives on risk factors for EHS that focus on hydration, heat acclimatization, medical conditions, climate change and policies, medications, and strength and conditioning sessions.

***Keywords**:* environment, exercise, heat acclimatization, heat tolerance, hydration

Exertional heat stroke (EHS) is a life-threatening condition characterized by an internal temperature ≥40°C with central-nervous-system dysfunction (Casa et al., 2015). EHS is interesting to study when examining prevention strategies and factors that predispose individuals to develop the condition. Many EHS-prevention strategies have been adopted to enhance exercise heat tolerance, but not necessarily to decrease the incidence of EHS (Alhadad, Tan, & Lee, 2019). Some examples of common strategies to enhance exercise heat tolerance include heat acclimatization (HA), hydration, work-to-rest ratios based on the environmental conditions, using body-cooling strategies, enhancing education, using prudence when dispensing supplements and medications that may contain substances that enhance metabolic rate or compromise thermoregulation, and modifying time or location of physical activity, among a myriad of other factors (Belval et al., 2018; Casa et al., 2015; Périard, Racinais, & Sawka, 2015; Rav-Acha, Hadad, Epstein, Heled, & Moran, 2004). Many of these key items are addressed in this review. Table 1 presents predisposing factors that may increase the risk of EHS. It is imperative to remember that the efforts to prevent EHS from occurring in the first place are dramatically different than the strategies that need to be in place to properly treat EHS if it occurs (Parsons, Anderson, Casa, & Hainline, 2019). Moreover, when the medical and scientific literature or mass media make note of an EHS case or death, we need to determine whether the system failure is related to EHS-prevention policies and procedures or it occurred with inadequate medical treatment (Parsons et al., 2019). This nuance is critical, because most EHS cases should never occur in the first place. Simple strategies related to enhanced fitness, HA, and work-to-rest ratios based on environmental conditions, as three examples, have been shown to be very effective in enhancing exercise heat tolerance.

The challenge of studying items that prevent EHS is that you cannot implement a strategy and then attempt to induce EHS, which would entail obvious ethical issues. In addition, given that the condition is so rare, it is often hard to capture enough cases in a given population to properly assess the influence of a prevention strategy. Our best opportunity is to examine the influence of internal and external risk factors for EHS using large-scale data sets from the military. The classic study by Rav-Acha et al. (2004) uniquely contributed to the EHS literature by collecting data on over 130 cases of EHS. It is rare for a data set this large to include both retrospective (what the EHS patient was doing before the incident) and prospective (what kind of recognition and treatment occurred after EHS) data. Although there are a few larger EHS data sets, the Rav-Acha data set stands alone in terms of the comprehensive case-specific data related to aspects of prevention, recognition, and treatment. Figure 1 presents an exploration of factors long suspected to play a role in incidence (adapted from Rav-Acha et al., 2004). For example, low physical fitness was present in a staggering 71% of the cases, and relatedly, overweight was present 64.7% of the time (Rav-Acha et al., 2004). These pieces of data seem striking but are supported by the recent 2019 meta-analysis by Alhadad et al. (2019) that was conducted to examine the efficacy of heat-mitigation strategies on core temperature and endurance exercise, which ranked aerobic fitness as the most important factor for protection in the heat (followed by HA, preexercise cooling, and hydration, respectively). The deep dive in this brief review will look to examine new developments for a few specific intrinsic and extrinsic factors. The Rav-Acha study showed that key factors like the aforementioned consistently play a role in the etiology of EHS, so they warrant our effort to better understand the role they have in exercise heat tolerance. Twelve different prevention strategies were present in more than 10% of the EHS cases (Rav-Acha et al., 2004), which certainly warrants our attention in developing effective strategies to reduce incidence of this potentially catastrophic condition.

Hydration

Sweating is a thermoregulatory mechanism that dissipates excess heat during exercise. Without adequate fluid intake, sweat loss may result in hypohydration (>2% body-mass loss) impairing athletic performance, cognitive performance, and physiological function (Belval et al., 2018; Casa et al., 2015; Lopez et al., 2011; Rav-Acha et al., 2004). Dehydration precipitated by sweat-based fluid loss is further exacerbated in hot and/or humid conditions (McDermott et al., 2017). Studies suggest that during exercise in the heat, for every 1% of body mass lost, internal temperature increases by approximately 0.15–0.22°C (Huggins, Martschinske, Applegate, Armstrong, & Casa, 2012; McDermott et al., 2017). Therefore, hypohydration must be limited to protect exercising individuals from exertional heat illnesses.

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Table 1 Risk Factors for Exertional Heat Stroke

Predisposing factor	Intrinsic	Extrinsic	Modifiable?
Environmental condition		✓	No
Heat policies		✓	Yes
Clothing		✓	Yes
Medical condition	✓		No
Previous heat complications	✓		No
Medication	✓	✓	No
Motivation	✓	✓	Yes
Recent illness	✓		No
Anthropometric characteristics	✓		No
Hydration	✓	✓	Yes
Supplements	✓		Yes
Exercise intensity/conditioning sessions	✓	✓	Yes
Heat acclimatization	✓		Yes
Fitness	✓		Yes

Adapted from Hosokawa, Y., Casa, D.J., Trtanj, J.M., Belval, L.N., Deuster, P.A., Giltz, S.M., . . . Jardine, J.F. (2019). Activity modification in heat: Critical assessment of guidelines across athletic, occupational, and military settings in the USA. *International Journal of Biometeorology, 63*(3), 405–427.

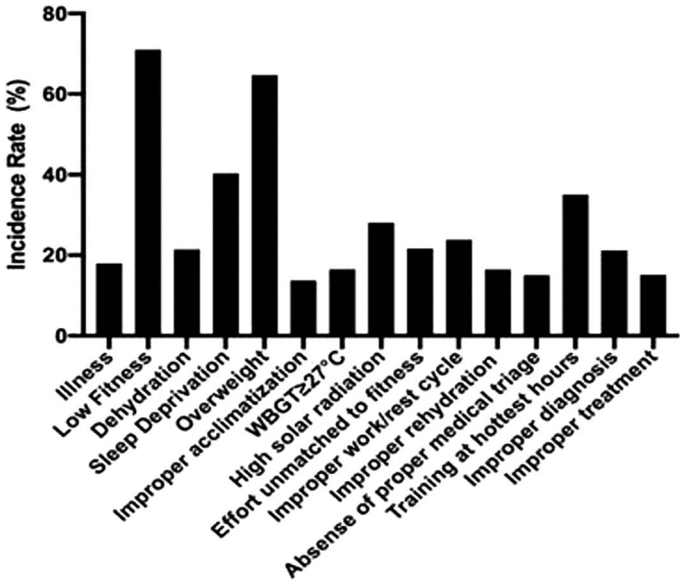

Figure 1 — Incidence rates for risk factors for exertional heat stroke in nonfatal cases. WBGT = wet-bulb globe thermometer. Adapted from Rav-Acha, M., Hadad, E., Epstein, Y., Heled, Y., and Moran, D.S. (2004). Fatal exertional heat stroke: A case series. *American Journal of the Medical Sciences, 328*(2), 84–87.

To mitigate risk of heat illness through hydration, appropriate hydration-assessment tools and fluid-replacement strategies must be implemented. Fluid-intake guidelines are highly debated among sport scientists and athletes, with large emphasis on two fluid-intake approaches: drinking to thirst/ad libitum intake and programmed and personalized fluid intake (Goulet & Hoffman, 2019; Kenefick, 2018). Sensation of thirst is thought to be an effective tool to avoid hypohydration as it is controlled biologically and mediated through osmoreceptors, baroreceptors (stimulated by plasma volume changes), and anti-diuretic hormone release (Cheuvront, Kenefick, Montain, & Sawka, 2010). Many claims in favor of drinking to thirst are influenced by safety concerns that arise from overconsuming fluids, which may occur during programmed drinking and may increase the risk of exercise-induced hyponatremia (McDermott et al., 2017). However, studies examining drink to thirst during endurance performance have reported greater thermoregulatory strain than with programmed drinking strategies (Beis, Wright-Whyte, Fudge, Noakes, & Pitsiladis, 2012; Daries, Noakes, & Dennis, 2000; Dion, Savoie, Asselin, Gariepy, & Goulet, 2013), indicating that sensation of thirst may not be sensitive enough to appropriately manage euhydration during exercise (Sawka et al., 2007). In addition, Adams, Vandermark, et al. (2016) reported that a large bolus of fluid consumed after exercise in a hypohydrated condition significantly reduced perception of thirst, even when body-mass loss exceeded 2%. Intake of the large bolus of fluid negated differences in thirst between the hypohydrated and euhydrated conditions (Adams, Vandermark, et al., 2016), indicating that it might be difficult to use thirst as a measure of hydration when consuming large boluses of fluid. Programmed drinking is a strategy customized for each athlete based on his or her sweat losses during exercise (Sawka et al., 2007). This strategy recommends that athletes consume fluids to keep sweat losses within 2% of body mass to avoid decrements in exercise and cognitive performance and protect against extreme fluid losses that could exacerbate heat strain (Kenefick, 2018). To maintain euhydration through programmed drinking, athletes must determine their sweat rate through acute changes in body mass. Bardis et al. (2017) compared a prescribed drinking strategy with ad libitum drinking during a cycling protocol that consisted of riding 30 km, with the last 5 km being an all-out performance assessment. Those authors reported that prescribed drinking provided a performance advantage (5.1% ± 4.8% faster speed) and significantly reduced internal body temperature and skin temperature postexercise (Bardis et al., 2017).

Although both methods have evidence to suggest they are appropriate strategies to mitigate hypohydration, there are multiple factors that must be considered prior to determining which strategy is optimal to reduce thermal strain. Fluid availability during an event/race, exercise duration, environmental conditions, and prehydration status are examples of factors that must be examined before determining a fluid strategy (Belval et al., 2018; Nuccio, Barnes, Carter, & Baker, 2017). For example, soccer games consist of two 45-min halves where drinking fluids during play is impossible, and programmed drinking before play and during halftime is the only hydration strategy that can be used maintain euhydration. In contrast, during cycling, fluid is always available, and drink to thirst may be an effective hydration strategy (depending on race duration and exercise intensity). Of note, despite the duration of exercise activity, athletes should always arrive at practices and games euhydrated (Belval et al., 2018). Furthermore, reducing body-mass loss during exercise through appropriate hydration strategies can help decrease the risk of developing an exertional heat illness.

Heat Acclimatization

Exercising in hot environments increases physiological strain by increasing core temperature, heart rate, and skin temperature and decreasing maximal oxygen consumption (Armstrong & Maresh, 1991; Nybo, Rasmussen, & Sawka, 2014). HA is a preventive strategy used in athletic, military, and industrial settings to improve

thermoregulation, cardiovascular stability, and tolerance of thermal strain (Périard et al., 2015). HA is defined as repeated heat exposure to condition the body to adapt to thermal loads. HA induction is rapid (begins on the first day of exposure), with most adaptations occurring within the first 4–7 days of exposure. HA induction periods can range from 5 to 21 days depending on the exercise intensity, duration, and frequency and number of heat exposures (Périard et al., 2015; Pryor, Johnson, Roberts, & Pryor, 2019). Most methods used to induce HA employ exercise modalities to increase thermal strain, but passive heat-exposure induction methods (e.g., sauna, hot bath, etc.) have been considered as an alternative to the exercise-induced method (Heathcote, Hassmén, Zhou, & Stevens, 2018). However, passive HA induction methods should not be used alone but, rather, in conjunction with exercise-induced HA to elicit a sufficient thermal impulse (Heathcote et al., 2018).

The objective of HA is to elicit sufficient thermal loads to improve physiological responses during exercise-induced heat stress (especially in hot and/or humid environments). HA has been shown to increase sweat rate, increase sweating threshold (begin sweating at a lower internal temperature), decrease sweat sodium and potassium concentrations, increase plasma volume, decrease resting and exercising heart rate, decrease the rate of rise in core temperature, decrease skin temperature, improve heat-shock protein response to thermal stress, improve thermal comfort/sensation, and improve cognition (Périard et al., 2015). The multitude of physiological adaptations resulting from proper HA decreases the risk of exertional heat illness (while improving exercise performance), and individuals not exposed to HA are at greater risk.

The positive adaptations from HA have led entities to implement HA guidelines in sport to protect athletes. For example, the National Collegiate Athletic Association (NCAA) and National Football League (NFL) have adopted HA guidelines (in 2003 and 2011, respectively) to prepare football players for hot and humid conditions during preseason. Before HA policies were put into effect in the NCAA, approximately one or two EHS-related deaths occurred in student-athletes during preseason or early season (August) each year. After the NCAA HA policies were implemented, EHS-related deaths decreased by 95% from 2003 to 2016 (Adams, Casa, et al., 2016). Similarly, the National Athletic Trainers' Association Inter-Association Task Force for Preseason Secondary School Athletics Participants (NATA-IATF) released preseason HA guidelines recommending that secondary-school athletes perform gradual HA during the first 2 weeks of preseason (Casa et al., 2009). A recent investigation examined the effectiveness of the NATA-IATF HA guidelines for reducing exertional heat illness and reported that state high school athletic association–mandated HA guidelines that met the NATA-IATF recommendations were associated with a 55% reduction in the incidence of exertional heat illness (Kerr et al., 2019). That study and others further reiterate the importance of HA guidelines and programs and their effectiveness in mitigating risk of EHS (Adams, Casa, et al., 2016; Kerr et al., 2019).

Medical Conditions

Obesity

Obesity is thought to influence thermoregulatory responses due to the specific heat of fat (2.97 $kJ \cdot kg^{-1} \cdot °C^{-1}$). The specific heat of fat is lower than that of lean mass (3.66 $kJ \cdot kg^{-1} \cdot °C^{-1}$), which indicates that a greater temperature change occurs for a given amount of heat energy and theoretically results in a higher core temperature in individuals with higher body-fat percentages (Geddes & Baker, 1967). A case-control study performed in soldiers training in hot and humid environments found that obese soldiers (BMI >27) were at a higher risk of heat illness, with an odds ratio (OR) of 3.53 (95% CI 1.90, 6.55). In addition, Bedno, Urban, Boivin, and Cowan (2014) examined risk of exertional heat illness during U.S. Army basic training in 9,455 men with 67 incidents of exertional heat illness occurred within 180 days. The risk of exertional heat illness was significantly greater in individuals who exceeded U.S. Army body-fat standards compared with normal weight (OR 3.98, 95% CI 2.17, 7.29). Dougherty, Chow, and Kenney (2010) found that in children, obese boys had a quicker inflection in internal temperature induced by changes in water-vapor pressure (above the threshold resulted in continuous rise in internal temperature) at several ambient temperatures than did lean boys. As the aforementioned studies indicate, there appears to be a link between body-fat percentage and risk for exertional heat illness. However, individuals with high body-fat percentage tend to have lower fitness levels, which has also been shown to increase the risk of exertional heat illness (Casa et al., 2015). Therefore, the effects of body fat and fitness as separate risk factors must be further investigated.

Spinal-Cord Injury

Spinal-cord injury can severely alter physiological function, which includes thermoregulation (Price, 2006). Injury to the spinal cord impairs motor function and reduces afferent signals to the hypothalamus, which controls thermoeffector responses that regulate body temperature (Price & Campbell, 1999; Price & Campbell, 2003; Price, 2006; Sawka, Latzka, & Pandolf, 1989). In environmental extremes such as heat, individuals with spinal-cord injuries have a reduction in both vasomotor control (skin vasodilation) and sweating capacity, which causes core temperature to rise at a faster rate than in non-spinal-injured individuals (Sawka et al., 1989). As heat storage increases, individuals with spinal-cord injury are unable to dissipate the amount of heat generated, which can increase their risk of heat-related illnesses (Hopman & Binkhorst, 1997). This is of particular interest in individuals with spinal-cord injuries who participate in wheelchair-bound sports. Participation in wheelchair-bound sports is growing rapidly, and many sporting venues such as the Paralympic games are hosted in hot and/or humid environments (Cooper, 1990). Therefore, athletes with spinal-cord injuries must be aware of the challenges of participating in sport in the heat and should participate in heat-acclimatization programs to improve their heat tolerance. Athletes should also note that the severity of thermoregulation impairment depends on the location of the spinal-cord injury, and individuals with spinal-cord lesions at T6 or higher are more likely to experience greater thermal strain and are at greater risk of EHS (Price, 2006).

Skin Conditions

Heat is dissipated from the body to its surrounding environment through the skin (Tansey & Johnson, 2015). There are several skin disorders that impair evaporation and skin vasodilation (thermoeffector pathways), which could increase risk of exertional heat illness (Austin, Hansbrough, Dore, Noordenbos, & Buono, 2003; Ben-Simchon, Tsur, Keren, Epstein, & Shapiro, 1981; Ganio et al., 2015; Leibowitz, Seidman, Laor, Shapiro, & Epstein, 1991;

Roskind, Petrofsky, Lind, & Paletta, 1978). Patients with third-degree burns covering 40–55% of their body surface area have been shown to be at greater risk for heat intolerance (Ganio et al., 2015). The burned areas of the skin are unable to produce sweat in the damaged area (replaced with split thickness grafts), which is problematic for heat dissipation, as evaporation of sweat is the most important avenue for heat loss. Although unburned areas are unaffected and able to produce sweat, there is still impaired heat dissipation, which results in hyperthermia (heat production exceeding heat loss). For example, Ganio et al. (2015) found that patients with burns covering over 40% of their body surface area had higher increase in core temperature (1.6°C ± 0.5°C) at the end of 90 min of exercise than patients with 17–40% of affected body surface area (1.2°C ± 0.3°C) and a control group (0.9°C ± 0.2°C). In that investigation, although whole-body sweat rate was similar between groups, the patients with 40% of their skin affected had significantly less functional skin available to secrete sweat, indicating that their sweat rate relative to functional skin was much higher (Ganio et al., 2015). In very hot and humid environments, this could increase the risk of developing a heat illness.

Other skin conditions such as psoriasis, schelorderma, and erthromelalgia have been linked to thermoregulatory dysfunction (Davis, Genebriera, Sandroni, & Fealey, 2006; Flavahan, 2008; Leibowitz et al., 1991). For example, Leibowitz et al. (1991) reported that males with psoriasis covering approximately 4.9% of their body surface area had significantly higher rectal temperatures at the end of a heat tolerance test than controls (38.3°C ± 0.1°C and 37.5°C ± 0.1°C, respectively). Although studies examining skin disorders and thermoregulatory function have reported that skin disorders may increase the risk of heat intolerance, more research is warranted to examine whether heat intolerance increases susceptibility to EHS.

Climate Change and Absence of Heat Policies

While exposure to hot and humid conditions has always posed a threat for exertional heat illnesses, the changing climate in response to human activities is increasing the likelihood of heat-related hazards. Average temperatures across the planet have increased approximately 1°C over the period from 1901 to 2016, and we are now in the warmest period in modern human history (Wuebbles et al., 2017). Record-breaking climate extremes are occurring with greater frequency, with 16 of the 17 warmest years on the instrumental record occurring from 2000 to 2016 (Wuebbles et al., 2017). It is expected that there will be a continuation of the pattern toward more frequent and severe extreme heat events across the globe (Pachauri et al., 2014, Perkins, 2015).

Recent heat waves that affected the 2019 Women's World Cup in France and forced suspension of several matches at the Australian Open are consistent with the pattern of climate change and likely to be an ongoing hazard for sporting events (Dobush, 2019). These events also highlight the importance of developing heat policies as a way to mitigate heat-related hazards. The cornerstone of such policies is a combination of weather-based activity modification and HA (see HA section; Hosokawa et al., 2019). Activity modification involves adjusting the length, duration, and/or intensity of activities based on weather conditions to reduce heat strain and the risk of exertional heat illness (e.g., Armstrong et al., 2007; Department of the Army and Air Force, 2013; American Conference of Governmental Industrial Hygienists, 2017). Many heat policies in athletic, occupational, and military settings rely on the wet-bulb globe temperature (WBGT), which integrates the influences of multiple factors that influence heat stress (e.g., sun, humidity, wind speed, and ambient air temperature; Yaglou & Minard, 1957).

The impact of heat-safety policies on exertional heat illness is well illustrated by a study of Georgia high school football players. Researchers in that study compared rates of exertional heat illness (heat syncope and heat exhaustion) in football players before and after the implementation of an evidence-based heat policy (Cooper et al., in press). The new policy required all the athletes to follow a 5-day acclimatization period where practices cannot last longer than 2 hr and players may wear no protective equipment except for a helmet and mouthpiece (Georgia High School Association, 2019). In addition, all schools must collect on-site WBGT measurements and follow WBGT-based activity-modification guidelines (Georgia High School Association, 2019). The study collected 6 years of data with 178,230 athlete exposures in the 6-year prepolicy period and 163,118 athlete exposures in the postpolicy period. Study results showed reductions of 35–100% in the rate of heat syncope/heat exhaustion when comparing the pre- and postpolicy periods, depending on the WBGT category (Figure 2). Furthermore, since the implementation of the policy in 2012, there have been no recorded heat-related deaths of Georgia football players when heat-policy guidelines were followed, which indicates that monitoring environmental conditions and following heat policies can significantly reduce the occurrence of heat illness.

Medications

Nonsteroidal Anti-Inflammatory Drugs

Nonsteroidal anti-inflammatory drugs (NSAIDs) increase gastrointestinal (GI) permeability and inflame the mucosa by reducing ATP production and prostaglandin synthesis via cyclooxygenase

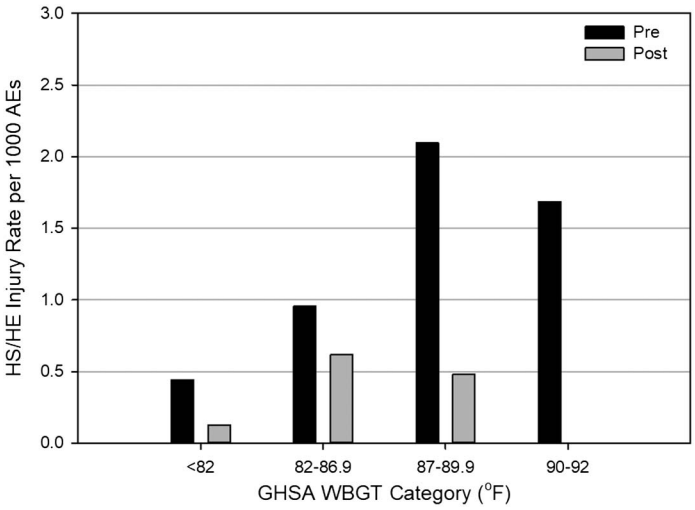

Figure 2 — Heat-syncope (HS) and heat-exhaustion (HE) injury rates by Georgia High School Association (GHSA) activity-modification category of pre- and postpolicy periods. AE = ; WBGT = wet-bulb globe thermometer. Reprinted with permission from Cooper, E., Grundstein, A., Miles, J., Curry, P., Casa, D., and Ferrara, M. (In press). The Georgia Study: The risk of exertional heat illness in high school football players. *Journal of Athletic Training*. © National Academy of Kinesiology and the American Kinesiology Association

inhibition in the GI mucosa. The increase in permeability may allow endotoxins (lipopolysaccharides [LPSs]) and other harmful materials to pass through the GI tract and into systemic circulation (Lambert, Broussard, Mason, Mauermann, & Gisolfi, 2001). LPSs overwhelm the liver, and the excess leaches into systemic circulation (endotoxemia), creating a systemic inflammatory response (Lim & Mackinnon, 2006). Intense and prolonged exercise has also been shown to increase GI permeability, with NSAIDs intensifying this effect (Ryan, Chang, & Gisolfi, 1996). This could potentially lead to an increased leaching of LPSs, enhancing the systemic inflammatory response. An additional side effect of NSAID use is splanchnic hypoperfusion, which can lead to further disruption of the GI tract (Guy & Vincent, 2018). Heat stress from the external environment, metabolic heat production, and the inflammatory response from the endotoxemia could place an individual at a higher risk for the development of heat stroke. Multiple rodent studies suggest that nonselective COX inhibitors (i.e., ibuprofen, indomethacin) increase the damaging effects of EHS on the body, decrease recovery time, and increase mortality (Audet et al., 2017). However, studies on NSAID use while exercising in the heat in humans are limited, and further research is warranted.

Stimulants

Stimulants (e.g., amphetamines, methylphenidate) were the second-most-detected banned substance reported by the World Anti-Doping Agency in 2013 (Zaretsky, Brown, Zaretskaia, Durant, & Rusyniak, 2014). Amphetamines may increase heat production through nonshivering thermogenesis in skeletal muscle, as well as decreasing heat dissipation through cutaneous vasoconstriction (Borbely, Baumann, & Waser, 1974; Sprague, Mallett, Rusyniak, & Mills, 2004; Zaretsky et al., 2014). Amphetamines and similar drugs (i.e., methylphenidate) are reported to mask the increase in time to exhaustion and increase internal temperature at exhaustion in warm environments (Nichols, 2014; Zaretsky et al., 2014). Methylphenidate did not change perceived exertion and thermal stress, which can place an individual at risk for a dangerous increase in internal temperature by inhibiting protective signals from the central nervous system to cease exercise (Lopez & Casa, 2009; Roelands et al., 2008). From 2006 through 2016, stimulant use doubled in the United States and its territories (Piper et al., 2018). Clinicians should be aware of athletes using daily stimulant medications, and specific recommendations and/or restrictions should be made for these athletes when they exercise in hot environmental conditions.

Other Medications and Supplements

The list of medications and supplements that may influence heat tolerance is extensive, and a few additional agents will be discussed. Aspirin has been shown to increase internal body temperature in subjects while exercising in the heat compared with a control (Bruning, Dahmus, Kenney, & Alexander, 2013). Antimuscarinic-like drugs (e.g., antihistamines, antipsychotics, tricyclic antidepressants) are known to affect heat dissipation and may predispose exercising individuals to EHS (Levine, LoVecchio, Ruha, Chu, & Roque, 2012; McAllen & Schwartz, 2010). Medications and other drugs known to influence hydration (e.g., diuretics, alcohol) can induce dehydration, which should be considered a risk for developing EHS (Hosokawa et al., 2019). Cardiovascular drugs (e.g., beta blockers, calcium channel blockers) can influence thermoregulation by inhibiting cutaneous vasodilation (Hosokawa et al., 2019; Nichols, 2014). Finally, common supplements such as creatine and caffeine have been thought to increase the risk of heat-related illness. However, the results from multiple studies have been inconclusive (Lopez & Casa, 2009; Nichols, 2014).

Strength and Conditioning Sessions

In the past 20 years (1998–2018), 35 football players died from either EHS or an underlying heart condition during practice or preseason conditioning workouts (Caterisano et al., 2019). The incidents typically occur when athletes are at a lower level of fitness and/or unacclimatized to environmental stressors when returning to training after a duration of inactivity from academic breaks (Caterisano et al., 2019). When athletes are exposed to intense training sessions after periods of inactivity, their risk of developing non-contact-related injuries (e.g., EHS) increases. Proper transition back to activity before being exposed to a heavy workload must occur to reduce the risk of EHS. Furthermore, strength and conditioning sessions must be introduced intentionally, gradually, and progressively to minimize risk of heat illness and support acclimatization (Parsons et al., 2019).

Exercise intensity is directly related to the amount of metabolic heat production (e.g., if intensity increases, metabolic heat production increases; Casa et al., 2015). It has been shown that intensity of exercise is a determinant for the rate of rise in rectal temperature and that those with higher sustained metabolic rates (i.e., increased exercise intensity) have a greater risk of heat injury (Casa et al., 2015; Noakes et al., 1991). Oftentimes individuals will modify their intensity as a protective mechanism to prevent reaching dangerous temperatures (Marino, 2004; Tucker & Noakes, 2009). However, influence from coaches, strength and conditioning staff, teammates, and self-motivation may suppress this mechanism and increase the risk of EHS (Casa et al., 2015). Occupational laborers may also increase their workload (i.e., intensity) to increase their financial compensation or to meet quotas without taking the necessary breaks (Hosokawa et al., 2019).

Untrained individuals are more susceptible to heat illness than trained individuals (Cleary, 2007). This can be problematic for athletes returning to training camp or conditioning sessions after a duration of inactivity, specifically in periods of hot and humid weather to which they are not acclimated. A strong predictor for thermoregulatory capability is cardiovascular fitness (Cleary, 2007; Gardner et al., 1996). Strength and conditioning coaches should be mindful of their athletes' physical fitness levels at the start of training to limit the risk of their developing an exertional heat illness. In addition, athletes should know their individual limits and not succumb to the influence of others to overexert and injure themselves. Specifically, freshman, transfer, and walk-on athletes may feel the need to work more to impress the coaching staff for a spot on the roster (Caterisano et al., 2019). Strength and conditioning coaches should educate these athletes to know their limits and gradually increase their workloads to avoid increasing their risk of heat illness.

Conclusion

Understanding factors that improve or impair exercise heat tolerance is critical to implementing preventive strategies to reduce the risk of EHS. EHS is 100% survivable with appropriate care, and it is imperative to identify factors that may influence the risk of this

condition. Health care professionals responsible for overseeing patients with heat-related emergencies must be properly educated on these risk factors. Implementation and education on these key factors will keep athletes, laborers, and war fighters safe and reduce the incidence of this potentially life-threatening condition.

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